



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR MATHEMATICS CLUB

PROBLEM NUMBER: WP2604

DATE: FEBRUARY 28, 2026

DUE DATE: MARCH 10, 2026

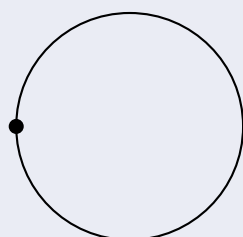
WEEKLY PROBLEM

EXERCISE. Given m points on a circle, join every pair with a chord such that no three chords intersect at a single interior point.

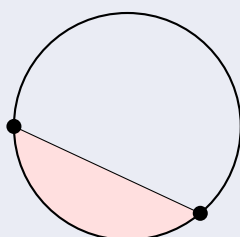
Question 1. If m points are placed on a circle and every pair of points is joined by a chord, how many chords are drawn?

Question 2. How many intersection points are formed inside the circle when no three chords are concurrent?

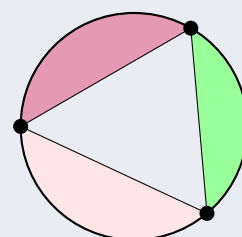
Question 3. Using the results from the above parts, obtain a formula for the maximum number of regions formed inside the circle.



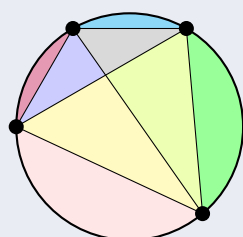
$$n = 1, c = 0, r_G = 1$$



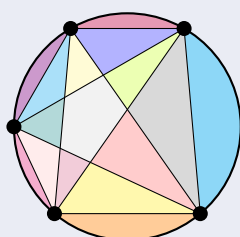
$$n = 2, c = 1, r_G = 2$$



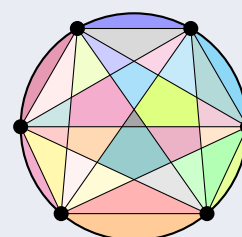
$$n = 3, c = 3, r_G = 4$$



$$n = 4, c = 6, r_G = 8$$



$$n = 5, c = 10, r_G = 16$$



$$n = 6, c = 15, r_G = 31$$



(Scan this to submit solution!)